Hands-On Lab (HOL): Ansible Setup on Ubuntu (AWS)

Lab Objective

Set up Ansible on Ubuntu (AWS EC2) with correct passwordless SSH configuration using AWS PEM key.

Lab Architecture

- 3 Machines (Ubuntu EC2)
- 1 Master (Ansible installed)
- 2 Clients (Managed Nodes)
- Same VPC / Network
- Use Private IPs

Step 1: Update System (All Machines)

```
sudo apt update -y
```

Step 2: Install Ansible (Master Only)

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y software-properties-common
sudo add-apt-repository --yes --update ppa:ansible/ansible
sudo apt install -y ansible
```

Verify:

```
ansible --version
```

Step 3: Configure Inventory

```
sudo vi /etc/ansible/hosts
```

Add:

```
[dev]
172.31.x.x ansible_user=ubuntu
172.31.x.x ansible_user=ubuntu
```

Step 4: Configure Ansible

```
sudo vi /etc/ansible/ansible.cfg
```

```
[defaults]
inventory = /etc/ansible/hosts
host_key_checking = False
remote_user = ubuntu
```

🚨 IMPORTANT: AWS SSH Reality

- AWS EC2 does NOT allow password-based SSH by default
- `ssh-copy-id` will FAIL if password login is disabled
- We MUST use PEM key to push public key

Step 5: Generate Key on Master

```
ssh-keygen
```

Creates:

- ~/.ssh/id_rsa (private)
- ~/.ssh/id_rsa.pub (public)

Step 6: Copy Public Key to Clients (CORRECT METHOD)

❌ Old Method (Not working in AWS):

```
ssh-copy-id ubuntu@client-ip
```

✅ Correct AWS Method:

```
cat ~/.ssh/id_rsa.pub | ssh -i /path/to/aws-key.pem ubuntu@<client-ip>
"mkdir -p ~/.ssh && cat >> ~/.ssh/authorized_keys"
```

Run for both clients.

Step 7: Fix Permissions (On Clients)

Login once using PEM:

```
ssh -i /path/to/aws-key.pem ubuntu@<client-ip>
```

Then run:

```
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

Step 8: Test Passwordless SSH

From Master:

```
ssh ubuntu@<client-ip>
```

✅ Should login without password or PEM

Step 9: Final Validation

```
ansible all -m ping
```

Expected:

```
SUCCESS => pong
```

🔥 Key Learning

Concept	Explanation
AWS PEM	Used only for initial login
id_rsa	Used by Ansible for automation
authorized_keys	Stores trusted public keys
